

31338 Network Servers

Lab 2a: Linux System Documentation

Aims

1. Use and configure the UNIX manual system
2. Locate installed package documentation, and documentation on the Internet

Task 1: Working with the Manual (Man) Pages

Begin by starting a separate terminal, `su` to root and run the following command in the background (with the "&"). This lab will need it later on, but it might take a while to run depending on your system.

```
mandb -cqs &
```

From another terminal, run the following 2 commands:

```
man 1 passwd
```

```
man 5 passwd
```

What is the difference between the two? (Record the answer in your learning journal). Next, run the command:

```
man passwd
```

Which one is shown (section 1 or section 5)?

Look in the file `/etc/man_db.conf`. Locate the `SECTION` entry in the file to verify the above behaviour. While you are in the file, also look at the `MANDATORY_MANPATH` section, to see in which directories the `man` command will locate man pages by default.

Next, run the following two commands and compare them (you may run them in two separate terminals):

```
man grep
```

```
info grep
```

Compare the `info` page to the `man` page (in general terms) and record notes in your learning journal. Are they basically the same, or are they quite different?

To conclude, check whether the `mandb` command has finished running (type: `jobs`, and see if it is listed). If not, wait for it to finish. The `mandb` command builds an index database of man pages. Normally it is run automatically overnight, but this activity requires us to run it manually because the virtual machine is never running overnight! Once it has finished running, try the following commands:

```
whatis passwd
```

```
apropos passwd
```

Which one returns more results? Why? (Hint: read the man pages for `"whatis"` and `"apropos"`).

Task 2: Finding Installed Package Documentation

Installed package documentation is in the directory `/usr/share/doc`. Browse the directory listing to see the available documentation (you do not need to write the list in your journal). To get a feel for what is available, go into the directory

```
/usr/share/doc/sed*/
```

This is documentation for the sed editor. The "*" will depend on the current version of sed installed on your virtual machine. Briefly scan the contents of this directory. You can view the contents of the FAQ file with:

```
zless sedfaq.txt.gz
```

(The `zless` command uncompresses the .gz file and pipe it to the "less" pager).

Task 3: Leveraging Online Resources

www.tldp.org (The Linux Documentation Project) is a good place to start for up-to-date Linux documentation. Use a web browser in the Host OS to carry out this task.

As an exercise in using the site, find the following items on www.tldp.org, and record the URL in your learning journal:

Find a HOWTO document about the Linux Clock Find the main Linux FAQ Find the Linux System Administrator's Guide (or SAG for short)

Also, try to use ChatGPT, Claude and Google Gemini to ask any questions.